Worksheet & Answer Key

When resistors are connected in series, the total resistance is the sum of individual resistor values. This means that each resistor must be sized to handle the current flowing through it without exceeding its maximum capacity. Adding more resistors in series increases the total resistance and reduces the overall current flow.

Word Bank:

current, resistors, series, values, total, increases, maximum, reduces

Fill in the blanks:

1. This means that each resistor must be sized to handle the _____ flowing through it without exceeding its maximum capacity.

2. When resistors are connected in _____, the total resistance is the sum of individual resistor values.

3. Adding more resistors in _____ increases the total resistance and reduces the overall _____ flow.

4. The total _____ is the sum of individual resistor _____.

5. This means that each resistor must be sized to handle the _____ flowing through it without exceeding its maximum _____.

6. The addition of more resistors in series does not affect the _____ value of a single resistor.

7. Adding more resistors in series _____ the total resistance and reduces the overall _____ flow.

8. The total _____ is reduced by adding more resistors in series.

9. Each resistor must be sized to handle the _____ flowing through it without exceeding its maximum _____.

■ Answer Key (for teachers only):

1. current
2. resistors
3. series
4. values
5. total
6. increases
7. maximum
8. reduces